

The Opportunity to Collect Kinetic Forces during a Round of Golf using Wearable Devices: Single-Subject Case Study – The Full Swings.

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ABSTRACT

This single-subject case study compared a golfer's swing kinematics and kinetics on a driving range and golf course using a pair of pressure sensor socks and a markerless motion capture solution. Results indicated a significant difference in the right and left plantar pressure percentages in wedge swings and 7-iron chest sway ($p < 0.05$). Findings suggest that wearable and mobile technology can measure discrete and longitudinal training and gameplay to enhance a golfer's performance.

Keywords: Wearable sensors, human pose estimation, kinetics, kinematics.

Index Terms: Image Processing and Computer Vision, Digitization and Image Capture; Visualization.

1 INTRODUCTION

The golf swing is a complex movement that requires dynamic balance and coordination of a golfer's upper and lower limbs and torso while controlling the trajectory of a golf club [1, 2]. Differences in dynamic balance have been reported to influence clubhead speed [3], are different in professional and amateur golfers [4], are unique to the individual [5], and change due to the topography of the golf course [6]. Maintenance of dynamic balance is important during changes in body posture [7], and it is measured using force plates, pressure mats, pressure sensor-embedded insoles, or socks [8, 9]. Mobile pressure sensors have been used for measuring dynamic balance during the golf swing and other sporting activities [3, 4, 10]. Driven by the interest to collect and measure human kinetic data during a game performance, the development of pressure sensor-embedded insoles and socks has created the ability to collect kinetic data during a round of golf and on the sports field.

On-course golf research using wearable pressure sensor technology is limited. Conventional golf research using force plates and pressure mats is generally conducted in a lab or studio and on a flat surface [5, 11, 12]. However, the topography of a golf course can be quite different from the flat surface of a golf practice range. The combination of uphill, downhill, and sidehill slopes can make the game challenging, requiring golfers to adapt their swing and dynamic balance based on a sloped playing environment versus flat ground [13]. In a study of 22 male professional golfers playing across 16 United Kingdom golf courses, 80% of the shots were played from sloped ground ranging between $\pm 1^\circ$ and 10° , with 50% of the shots played from

an incline greater than gradient slope of 2.25° [14, 15]. In a study examining GRFs on slopes, Blenkinsop et al. [14] created a 5° slope using a computer-assisted rehabilitation environment. They found that the golfer's center of pressure shifted towards the lower foot, and the golfers adopted a wider stance. Aside from the plantar pressure differences on uneven lies, different pressure patterns exist using golf clubs of different lengths [16, 17]. For example, golfers with a handicap of 0 – 14 achieved higher GRFs when hitting a 3-iron versus a driver on natural grass [16]. In addition, unique styles of plantar pressure tendencies were reported in a study conducted by Shepherd et al. [5] with 106 golfers; the results indicated that there is not a single best golf swing GRF method, sliding, jumping, or twisting/spinning, that was indicative of a primary force production mechanism and that some of the golfers adopted multiple styles based on the golf club used. Styles were based on tour player GRF means derived from a proprietary software app (Swing Catalyst) categorizing golfers with higher horizontal GRF as sliders which indicate a forward/backward motion with a mean of 19% body weight, and one standard deviation, golfers as jumpers who produce vertical GRF with a mean of 179% body weight and one standard deviation, and golfers who produce torque or twisting/spinning with a mean of 127.7 Nm and within one standard deviation [5]. Furthermore, a study of ten female collegiate varsity golfers performing four different shots with three golf clubs also reported individualized pressure patterns. The findings resulted in three classified patterns: (a) target-leg-weighted and the late peak force near the ball impact, (b) target-leg-weighted and early peak force, and (c) rear-leg-weighted and no peak force [18].

These findings suggest that golfers may use unique golf swing plantar pressure strategies based on different topographies of the course, such as limited pressure shifting on uphill shots, individual patterns based on the club used, and individualized styles to create clubhead speeds, whether trail or lead leg GRFs are used. Thus, this study aims to evaluate and introduce wearable plantar pressure and mobile technology to capture golf swing kinematics and plantar pressure during a round of golf and on the driving range.

2 MATERIALS AND METHODS

2.1 Study Design, Instrumentation, and Subject Preparation

This single-subject case study was designed to use wearable sensors and mobile technology to capture kinematic and kinetic data on a driving range and golf course. The authors' Institutional

Review Board approved this study. The subject was a 50-year-old male, a right-handed golfer with a zero handicap. The subject used his golf clubs, driver, 7-iron, and wedge during range swings and on the golf course. Data was collected on the range and the golf course using an Apple iPad Pro (Cupertino, CA, USA) set up at approximately 36" high and 8' to 12' from the subject, recording the golf swings at 240 frames per second. An Apple iPhone 13 was used for pressure data collection, and an iPad 2nd Generation was used for golf club and ball data.

Kinematic pelvis and chest sway data of the golfer's full swing was determined using the SportsBox AI (Bellevue, WA, USA) mobile app (ver. 1.8.0-4), and each 2D video was uploaded and converted to 3D kinematic swing metrics.

A pair of sensor-embedded Sensoria (Redmond, WA, USA) socks captured changes in pressure at 100 Hz during full swings. Each Sensoria sock comprises three proprietary piezoelectric sensors, one aligned under the calcaneus, a second under the first metatarsal phalangeal joint, and a third under the fifth metatarsal phalangeal joint. The sensors were not load-calibrated by the manufacturer for testing. The three-sensor sock transmitted data via a 7-gram Sensoria Core microelectronics module containing a MEMS 9-axis accelerometer, gyroscope, and magnetometer, Bluetooth Smart (low energy) technology, and powered by the Sensoria Core battery. The Sensoria Lab mobile app (ver. 4.4) captured changes in sensor data during the golf swings using an Apple iPhone 13.

2.2 Experimental Procedures

The subject put on the foot-specific Sensoria socks, attached the modules that automatically powered on, and proceeded to put on his regular golf shoes. The range session started after a self-selected warm-up. Before each swing, the subject was verbally given a countdown of three, two, one, go; the golfer then raised his heels with 100% of the pressure onto his forefoot and toes to create a synchronization event of the textile sensor-based sock data with the video. The subject then performed five full swings using his driver, 7-iron, and wedge. Researchers collected 2D golf swing video using the iPad Pro, plantar pressure changes using textile sensor embedded socks using the Sensoria Lab app. This completed the range session.

The researchers and the subject then headed to the golf course, where the subject played three golf holes: a par five, a par three, and a par four. Researchers collected 2D golf swing video using

the iPad Pro, plantar pressure changes using textile sensor embedded socks using the Sensoria Lab app. The driver, 7-iron, and wedge golf swings performed on the golf course were used for analysis.

3 DATA ANALYSIS

Sensor data captured during each swing was used to determine changes in right and left foot pressure and right and left foot heel and forefoot pressures. Sensors under the first and fifth metatarsals were combined to determine forefoot pressures. Sensor pressure data was determined using a min-max normalization during the execution of the start and stop of each full swing. Calculated changes in percentages of right and left heel and forefoot, and right and left feet plantar pressures were matched to key full swing positions (Figure 1).

Video frames captured at 240 Hz were converted to 100 Hz using Adobe Premiere Pro (San Jose, CA, USA) and synchronized with Sensoria pressure sensor data based on lifting heel event frames. Seven discrete key golf swing video frames were identified using Adobe's Premiere Pro software, including—1. Address (The address was the first frame before the club's movement into the backswing.), 2. Backswing Lead Arm Parallel to the Ground (lead arm parallel was determined when the golfer's left elbow joint and forearm were horizontal with the ground), 3. Top of Backswing (top of the backswing frame was determined when the golf club changed directions from backswing to downswing), 4. Downswing Lead Arm Parallel to the Ground (lead arm parallel was determined when the golfer's left elbow joint and forearm were horizontal with the ground), 5. Impact (Impact was determined when the club initially contacted the ball.), 6. Trail Arm Parallel to the Ground (trail arm parallel was determined when the golfer's right elbow joint and forearm were horizontal with the ground), and 7. Finish (The finish was determined when the golf club and body stopped moving in a golf swing pattern)—were identified for data analysis (Figure 1).

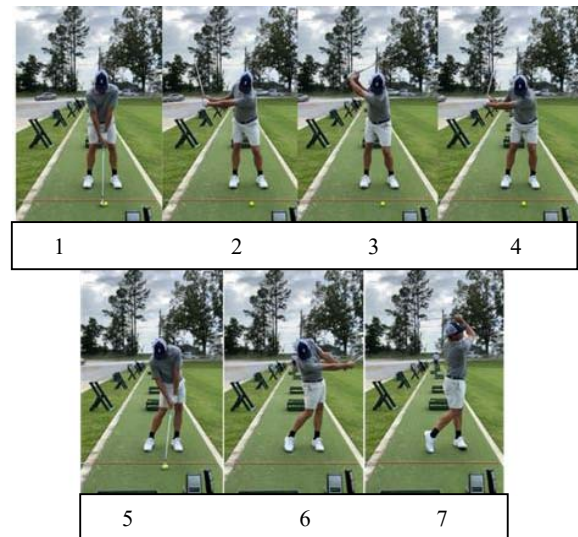


Figure 1: Seven key positions of the full swing: 1. Address, 2. Half-way Backswing, 3. Top of Swing, 4. Halfway Downswing, 5. Impact, 6. Halfway Follow Through, and 7. Finish.

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4 STATISTICAL ANALYSIS

Due to a single-subject design and the limited number of swings performed on the golf course compared to the range, this statistical analysis has low statistical power, and concerns of non-normality and violation of the assumption of independence of data were considered before conducting analysis, including generality of conclusions [19]. Kinetic and kinematic data were wrangled using Microsoft Excel (Redmond, WA, USA). A paired t-test was conducted using IBM SPSS ver. 28.0 (New York, NY, USA) to compare golf course and range conditions on plantar pressures and sway changes with an α -level set at 0.05. A two-tailed paired samples t-test determined a significant difference ($p=0.013$) in the total right and left plantar pressure percentages during the wedge swing on the course compared to the range and with 7-iron chest sway ($p=0.002$) (Table 1) and Cohen's d test resulted in less than small (0.2) to medium (0.5) effect size [20]. No other significant differences were found between the golf course and range swings. Figure 2 presents the differences in plantar pressure of the seven key full-swing positions on the range and on-course golf swing(s) for the wedge. Figure 3 presents the amount of pelvis and chest sway at the seven keyframes on the range compared to the golf course swing(s) for the wedge swing comparison.

Table 1. Results of Paired T-Test

Paired Sample Test			
	Mean	Std. Deviation	P-Value
7-iron Chest Sway: G.C. – Range	.9229	.4871	.002*
Wedge Left Foot % G.C. – % Range	-.2843	.2169	.013*
Wedge Right Foot % G.C. – % Range	.2843	.2169	.013*



Figure 2: Wedge Plantar Pressure Distributions at the seven key positions of the full swing: 1. Address, 2. Halfway Backswing, 3. Top of Swing, 4. Halfway Downswing, 5. Impact, 6. Halfway Follow Through, and 7. Finish.

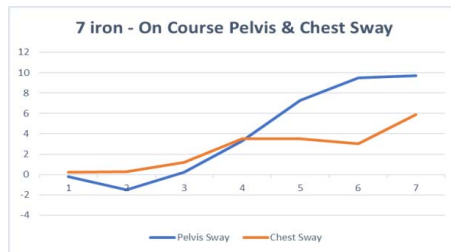


Figure 3: 7 iron Pelvis and Chest sway in inches at the seven key positions of the full swing: 1. Address, 2. Half-way Backswing, 3. Top of Swing, 4. Half-way Downswing, 5. Impact, 6. Halfway Follow Through, and 7. Finish.

5 DISCUSSION

This study aimed to evaluate and introduce wearable plantar pressure and mobile technology to capture golf swing kinematics and plantar pressure during a round of golf and on the driving range. Preliminary data suggests a significant difference in golf course and range right and left foot percentage change in plantar pressure with the wedge swings and the amount of chest sway with the 7-iron. No other significant differences were found between the golf course and range swings.

The significant difference in the golf course wedge shot could be due to the sloping fairway that created an uneven playing surface; the ball was below his feet compared to level ground, as



altered patterns were observed in uphill and downhill sloped fairways [14]. Evaluating swing kinetic performance, the subject moves or begins his backswing by laterally moving towards his right foot from Position 1 (address) into Position 2 (left arm parallel) when the golfer shifts towards his left foot while still making the backswing. Pressure shifts to the golfer's left foot as the downswing continues through impact and into the finish. This contrasts the GRF profiles of nine Korean professional golfers, which showed no significant changes in vertical forces until the top of the backswing (Position 3) [11]. In a follow-up discussion with the subject, researchers learned the subject had been trained to shift his weight from his right foot to his left foot as soon as possible when making his backswing, resulting in the early change in pressure from Position 2 to Position 3 before the club moved to the top of the swing.

5.1 Practical Application and Related Work

This case study provides the groundwork for extending research using wearable sensors and mobile technology to explore 3D kinematics and kinetics during play and practice. Determining kinematic and kinetic factors during gameplay has many benefits, including knowledge transfer and minimizing the risk of injury, which should interest golfers and golf instructors. Learning gameplay performance during play has been shown to increase knowledge transfer and improve the playing experience [21], which may be optimized using mobile technology to extend reality by displaying gameplay performance as this has been shown to promote positive learning experiences [22]. In addition, previous research articles have identified that augmenting computer graphics into reality can improve long-term memory retention, physical task performance, and skill training in naturalistic scenarios [23-25]. For instance, an on-course tracking system of a golfer's swing and ball flight tendencies can offer visually augmented aim targets overlayed on the course the golfer is playing to improve shot outcomes. Golf course topographies, unseen hazards, and swing guidance information, i.e., artificial intelligence coach or caddie, can be implemented via wearable and mobile technology.

Another benefit of collecting and quantifying gameplay data is minimizing golf swing-related injuries. Measuring upper and lower body segment flexion can be used to minimize the adverse physical stresses of the golf swing [26]. For instance, increases in lateral tilt (side bending or separation of chest and pelvis sway) are known to increase lumbar spine loads [27, 28]. In addition, Ferdinands et al. [29] identified that the downswing mechanics appear more critical due to the lateral shear stresses placed on lumbar spine vertebrae during the latter half of the downswing. In this study, the golfer's kinematic differences in the chest and pelvic sway will need further research to understand the significant difference in 7-iron chest sway (2.5" to 1.59") between the range and the on-course swing.

6 CONCLUSION

This single-subject case study highlights the opportunity to capture golf swing kinematics and ground reaction pressure timing patterns during on-range practice and on-course play using

wearable pressure and mobile technology; however, with a single subject, findings should not be over-generalized.

AR mobile and wearable technology can improve the delivery of on-course and on-range swing training and golf course playing information. The adoption of wearable sensor technology shows promise in tracking a golfer's performance over time, on the range, and in unique golf course topographies. Instructors and golfers can use kinetic and kinematic data to change full-swing and short-game movement patterns, identify movement patterns that may cause injuries, and serve as baseline timing for optimal performance. Furthermore, this study examined changes in plantar pressures.

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